TM Forum Component

Location Management

TMFC014

Maturity Level: Beta	**Team Approved Date: 18-Aug-2023**
Release Status: Roadmap	**Approval Status: Not yet approved**
Version 1.3.1	**IPR Mode: RAND**

Notice

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Table of Contents

TMFC014 – Location Management

1. Overview

Component Name	ID	Description	ODA Function Block
Location Management	TMFC014	The Location Management component allows easy reference to geographic places important to other entities, where a geographic place is an entity that can answer the question "where?" This component could be a facade tool for GIS systems, address sytems, BIM systems, mapping systems ... (eg. Google Maps).	Production

2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM L2 - SID ABEs links

(no eTOM business activities are listed for this component in the YAML — see 2.2)

2.2. eTOM business activities

eTOM business activities this ODA Component is responsible for:

(none listed in the component YAML)

2.3. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE L1	SID ABE L1 Definition	SID ABE L2 (or set of BEs)	SID ABE L2 Definition

Location	The Location ABE represents the site or position of something, such as a customer's address, the site of equipment where there is a fault and where is the nearest person who could repair the equipment, and so forth. Locations can take the form of coordinates and/or addresses and/or physical representations. Disclaimer This Location model is complex and has a number of subtleties. It provides a high level framework, explains broad concepts and gives illustrative examples. Actual solutions for particular requirements will be dependent factors such as country address standards state/ country / province map standards Service Provider addressing standards Equipment vendor equipment naming & addressing conventions whether textual only or graphical representations will be provided It is expected that when this model is used in NGOSS Catalyst projects, the feedback will clarify its application & usage in particular situations. There is obviously a limit to the amount of detail that can be included in a document like this. Those wishing to delve into more detail should consult the references used as the basis for this document. Introduction At the most fundamental level [Zachman], [Coad-Archetypes] the world can be thought of as comprising of entities that answer the 5 fundamental questions. (see below Figure L.00 - Showing how Location Fits in)	Geographic Place	A group of entities that are used to define a geographic location and enable it to be marked it on a map or given an address.

Location	The Location ABE represents the site or position of something, such as a customer's address, the site of equipment where there is a fault and where is the nearest person who could repair the equipment, and so forth. Locations can take the form of coordinates and/or addresses and/or physical representations. Disclaimer This Location model is complex and has a number of subtleties. It provides a high level framework, explains broad concepts and gives illustrative examples. Actual solutions for particular requirements will be dependent factors such as country address standards state/ country / province map standards Service Provider addressing standards Equipment vendor equipment naming & addressing conventions whether textual only or graphical representations will be provided It is expected that when this model is used in NGOSS Catalyst projects, the feedback will clarify its application & usage in particular situations. There is obviously a limit to the amount of detail that can be included in a document like this. Those wishing to delve into more detail should consult the references used as the basis for this document. Introduction At the most fundamental level [Zachman], [Coad-Archetypes] the world can be thought of as comprising of entities that answer the 5 fundamental questions. (see below Figure L.00 - Showing how Location Fits in)	Geographic Location	*(no description available)*

Location	The Location ABE represents the site or position of something, such as a customer's address, the site of equipment where there is a fault and where is the nearest person who could repair the equipment, and so forth. Locations can take the form of coordinates and/or addresses and/or physical representations. Disclaimer This Location model is complex and has a number of subtleties. It provides a high level framework, explains broad concepts and gives illustrative examples. Actual solutions for particular requirements will be dependent factors such as country address standards state/ country / province map standards Service Provider addressing standards Equipment vendor equipment naming & addressing conventions whether textual only or graphical representations will be provided It is expected that when this model is used in NGOSS Catalyst projects, the feedback will clarify its application & usage in particular situations. There is obviously a limit to the amount of detail that can be included in a document like this. Those wishing to delve into more detail should consult the references used as the basis for this document. Introduction At the most fundamental level [Zachman], [Coad-Archetypes] the world can be thought of as comprising of entities that answer the 5 fundamental questions. (see below Figure L.00 - Showing how Location Fits in)	Geographic Address	A group of entities used to represent the geographic address of a location.

Location	The Location ABE represents the site or position of something, such as a customer's address, the site of equipment where there is a fault and where is the nearest person who could repair the equipment, and so forth. Locations can take the form of coordinates and/or addresses and/or physical representations. Disclaimer This Location model is complex and has a number of subtleties. It provides a high level framework, explains broad concepts and gives illustrative examples. Actual solutions for particular requirements will be dependent factors such as country address standards state/ country / province map standards Service Provider addressing standards Equipment vendor equipment naming & addressing conventions whether textual only or graphical representations will be provided It is expected that when this model is used in NGOSS Catalyst projects, the feedback will clarify its application & usage in particular situations. There is obviously a limit to the amount of detail that can be included in a document like this. Those wishing to delve into more detail should consult the references used as the basis for this document. Introduction At the most fundamental level [Zachman], [Coad-Archetypes] the world can be thought of as comprising of entities that answer the 5 fundamental questions. (see below Figure L.00 - Showing how Location Fits in)	Geographic Site	*(no description available)*

Location	The Location ABE represents the site or position of something, such as a customer's address, the site of equipment where there is a fault and where is the nearest person who could repair the equipment, and so forth. Locations can take the form of coordinates and/or addresses and/or physical representations. Disclaimer This Location model is complex and has a number of subtleties. It provides a high level framework, explains broad concepts and gives illustrative examples. Actual solutions for particular requirements will be dependent factors such as country address standards state/ country / province map standards Service Provider addressing standards Equipment vendor equipment naming & addressing conventions whether textual only or graphical representations will be provided It is expected that when this model is used in NGOSS Catalyst projects, the feedback will clarify its application & usage in particular situations. There is obviously a limit to the amount of detail that can be included in a document like this. Those wishing to delve into more detail should consult the references used as the basis for this document. Introduction At the most fundamental level [Zachman], [Coad-Archetypes] the world can be thought of as comprising of entities that answer the 5 fundamental questions. (see below Figure L.00 - Showing how Location Fits in)	Local Place	A set of entities that determine where things are in relation to a local coordinate system. It provides a linking point to other parts of the SID model.

2.4. Functional Framework Functions

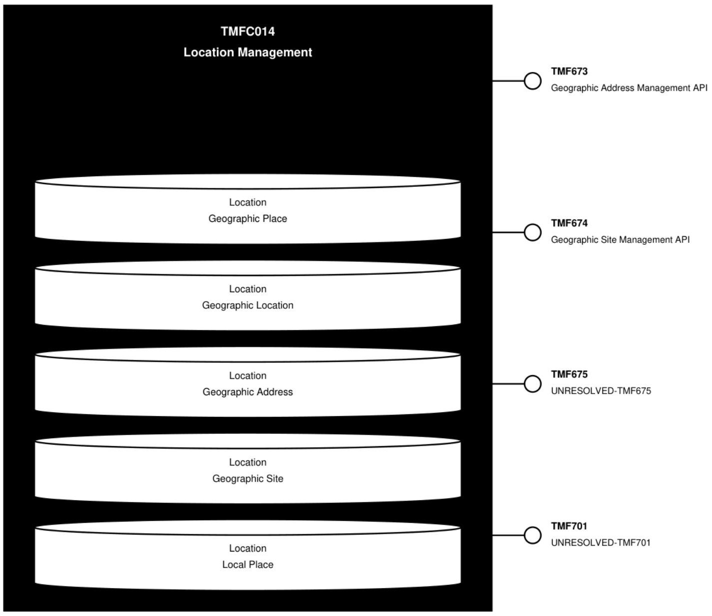
Function ID	Function Name	Aggregate Function Level 1	Aggregate Function Level 2	Function Description
429	Location Change History Management	Resource Management	Location Management	Location Change History Management; Tracks all changes of location data, making available attributes according to their historical values in certain periods.
430	PreFormattedLocation InformationPresentation	Resource Management	Location Management	Pre-formatted Location Information Presentation generates different views for different business cases (e.g. different format of address strings)
431	Location Information Updating	Resource Management	Location Management	Location Information Updating provides means to update the repository with new/updated location information from external sources.
432	Location Information Searching	Resource Management	Location Management	Location Information Searching provide the ability to search for a provided location/address, as part of the Location Management, including the ability to return near matches if an exact match is not found.

433	Location Structure Data Configuration	Resource Management	Location Management	Location Structure Data Configuration provides facilities for creating, modifying and deleting location structures data according to business rules of Service Providers or national and international location regulations. Also, utilities for defining sets of location attributes, levels and hierarchies should be available.
434	Location Data Integrity Management	Resource Management	Location Management	Location Data Integrity Management provides ability to maintain data integrity in the whole location repository. It's especially important if there are many external data sources that deliver new addresses for the repository.

3. TM Forum Open APIs & Events

The following part covers the APIs and Events; this part is split in 3: - List of Exposed APIs - This is the list of APIs available from this component. - List of Dependent APIs - In order to satisfy the provided API, the component could require the usage of this set of required APIs. - List of Events (generated & consumed) - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. API Context Diagram

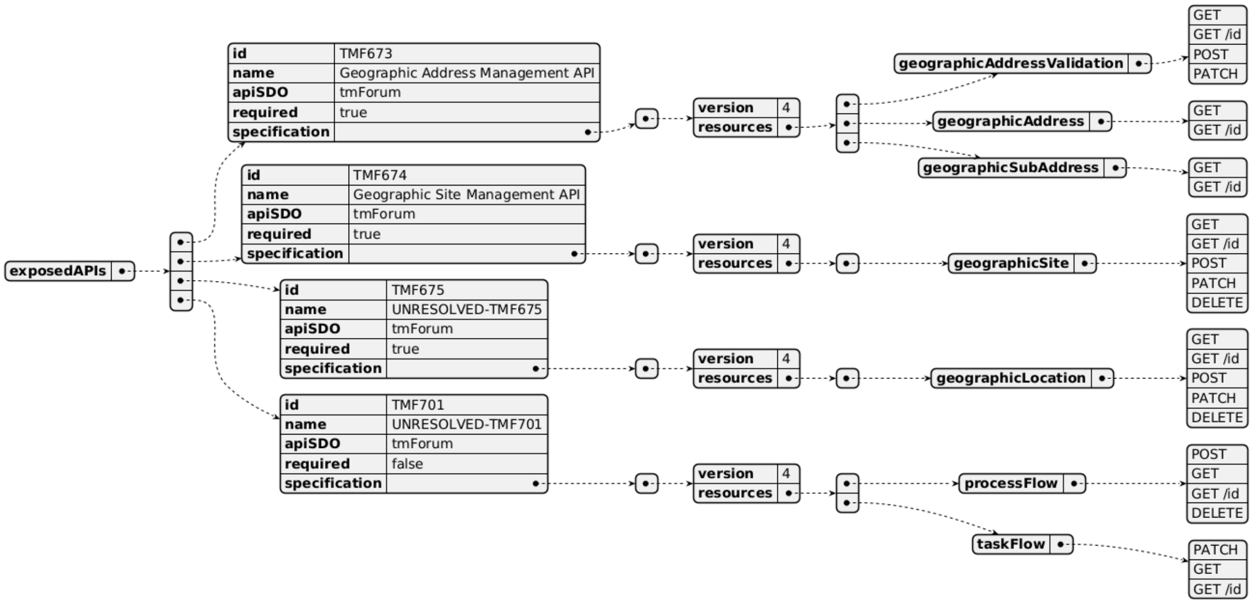


(SVG source: [TMFC014_API_Context.svg]() — hand-drawn rather than PlantUML for this one diagram, since PlantUML's automatic layout couldn't give each API its own straight, individually-anchored connector at a chosen height. Generated by the `component-specification-documentation` skill's `scripts/render_api_context_svg.py`.)

3.2. Exposed APIs

API ID	API Name	Mandatory / Optional	API Version	Resource	Operations
TMF673	Geographic Address Management API	Mandatory	4	geographicAddressValidation	GET, GET /id, POST, PATCH
TMF673	Geographic Address Management API	Mandatory	4	geographicAddress	GET, GET /id
TMF673	Geographic Address Management API	Mandatory	4	geographicSubAddress	GET, GET /id
TMF674	Geographic Site Management API	Mandatory	4	geographicSite	GET, GET /id, POST, PATCH, DELETE
TMF675	UNRESOLVED-TMF675	Mandatory	4	geographicLocation	GET, GET /id, POST, PATCH, DELETE

TMF701	UNRESOLVED-TMF701	Optional	4	processFlow	POST, GET, GET /id, DELETE
TMF701	UNRESOLVED-TMF701	Optional	4	taskFlow	PATCH, GET, GET /id



3.3. Dependent APIs

API ID	API Name	API Version	Mandatory / Optional	Resource	Operations

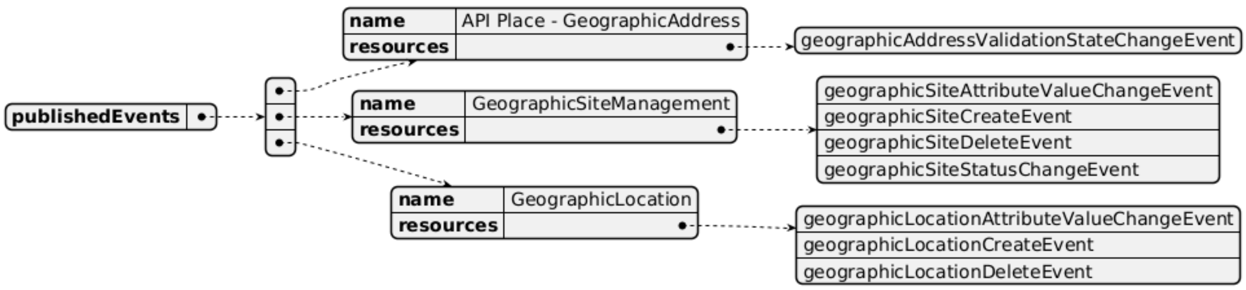
3.4. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.

Published Events

API ID	API Name	Event Resources
TMF673	Geographic Address Management API	geographicAddressValidationStateChangeEvent

TMF674	Geographic Site Management API	geographicSiteAttributeValueChangeEvent geographicSiteCreateEvent geographicSiteDeleteEvent geographicSiteStatusChangeEvent
TMF675	UNRESOLVED-TMF675	geographicLocationAttributeValueChangeEvent geographicLocationCreateEvent geographicLocationDeleteEvent



Subscribed Events

(none listed in the component YAML)

4. Machine Readable Component Specification

Refer to the ODA Component Directory on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
eTOM	*(none)*
SID	v23.5
Functional Framework	v23.0

5.2. Jira References

eTOM - ISA-559 Enrich eTOM with business activities necessary to manage any type of Location

5.3. Further resources

n/a

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date	Modified by	Description of changes
0.1.0	28-Jun-2022	Gaetano Biancardi, Sylvie Demarest	Final edits prior to publication
0.2.0	19-Aug-2022	Anastasios Sarantis	Changes asked from the community. Component description changed according to feedback + Functional Framework updates
1.0.0	07-Dec-2022	Goutham Babu	Initial release of document
1.0.1	25-Jul-2023	Ian Turkington	No content changed, simply a layout change to match template 3. Separated the YAML files to a managed repository.
1.0.1	15-Aug-2023	Amaia White	Final edits prior to publication
1.1.0	13-May-2024	Hugo Vaughan, Gaetano Biancardi	Aligned to Frameworks 23.5; aligned to latest template; removed TMF688 from exposed and dependent API
1.1.0	12-Jul-2024	Amaia White	Final edits prior to publication
1.2.0	24-Oct-2024	Gaetano Biancardi	Exposed API: fixed duplication of TMF674

1.2.0	14-Mar-2025	Rosie Wilson	Updated Maturity Level to Beta as confirmed by Gaetano Biancardi, as v4 of TMF675 is still in preview and API confirmed as Mandatory for this component by Anastasios Sarantis; final administrative updates

6.1.2. Release History

Release Status	Date Modified	Modified by	Description of changes
Pre-production	07-Dec-2023	Goutham Babu	Initial release of document
Pre-production	23-Jan-2023	Adrienne Walcott	Updated to member evaluation status
Pre-production	15-Aug-2023	Amaia White	New release 1.0.1
Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	12-Jul-2024	Amaia White	New release 1.1.0
Production	30-Aug-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	14-Mar-2025	Rosie Wilson	New release 1.2.0
Production	09-May-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status

6.2. Acknowledgements

Team Member	Company	Role
Anastasios Sarantis	Vodafone	Editor
Sylvie Demarest	Orange	Reviewer
Ian Turkington	TM Forum	Additional Input
Hugo Vaughan (TM Forum)	TM Forum	Additional Input